FinPlan-LLM: A Multi-Agent Financial Advisory System

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Abstract

Traditional large language models, when directly prompted for financial advice, produce plausible-sounding but fundamentally ungrounded recommendations. They cannot verify current market conditions, validate constraint compliance, or provide quantitative performance metrics. This paper presents FinPlan-LLM, a multi-agent financial advisory system built on a core design principle: *LLMs for reasoning and soft decisions, heuristics for hard financial decisions*. The system implements an 8-node LangGraph workflow using Google Gemini 2.5 Flash, featuring autonomous ETF research via a ReAct agent, real-time macroeconomic regime detection using FRED API data, systematic constraint enforcement through metadata filtering, deterministic portfolio allocation heuristics, historical backtesting with yFinance data, and a self-validation loop with LLM-as-Judge. Compared against direct prompting of Gemini 2.5 Flash, GPT-4o, and Claude Sonnet 4, FinPlan-LLM demonstrates superior performance through verifiable backtest metrics (Sharpe ratios of 1.03 to 1.93 versus unsubstantiated baseline estimates), systematic constraint satisfaction (filtering 4 to 8 ETFs based on user requirements versus verbal acknowledgment only), and three-layer explainability (console logs, PDF reports, and LangSmith traces). The system processes end-to-end requests in 60 to 110 seconds at approximately $0.38 per query, making sophisticated financial planning accessible while maintaining full transparency.

1 Introduction

1.1 Motivation

Large language models have demonstrated remarkable capabilities across diverse domains, yet their application to financial advisory presents fundamental challenges. When users request portfolio recommendations from general-purpose LLMs, they receive responses that appear professional and well-reasoned but suffer from critical limitations: the advice is grounded in training data that may be months or years out of date, constraints are acknowledged verbally but cannot be systematically verified, and performance estimates are plausible-sounding guesses rather than computed metrics.

Consider a user requesting an ESG-focused portfolio excluding oil companies. A vanilla LLM might recommend specific ETFs claiming they "screen out fossil fuels," but it cannot verify whether those funds currently hold oil-related positions. It cannot check today's Federal Reserve interest rate to inform allocation decisions. It cannot compute the actual Sharpe ratio of its recommended portfolio over the past year. These limitations represent not minor inconveniences but fundamental gaps that undermine the utility of LLM-based financial advice.

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1.2 Problem Statement

This project addresses the challenge of building an agentic AI system for personalized financial planning that:

1. Accepts natural language investment constraints and systematically enforces them
2. Integrates real-time market and economic data into decision-making
3. Provides quantitative validation through historical backtesting
4. Self-validates recommendations before presenting them to users
5. Offers comprehensive explainability at multiple levels of detail

1.3 Key Design Principle

The system architecture embodies a core principle that emerged through iterative development: **LLMs for reasoning and soft decisions, heuristics for hard financial decisions**. This hybrid approach recognizes that LLMs excel at understanding user intent, synthesizing research findings, and generating explanations, while deterministic algorithms provide the reliability required for quantitative calculations where errors could have real financial consequences.

For example, the system uses an LLM to interpret that "no oil companies" means filtering for fossil fuel exclusions, but uses a deterministic heuristic (100 minus age) to calculate base equity allocation. The LLM reasons about which ETFs best match a user's goals, but a mathematical formula computes the exact Sharpe ratio.

1.4 Contributions

This work makes the following contributions:

1. **Multi-Agent Architecture**: An 8-node LangGraph workflow that orchestrates specialized components for constraint parsing, macro analysis, autonomous research, allocation, backtesting, news analysis, validation, and reporting.
2. **Autonomous Research Agent**: A ReAct-based agent with four tools that conducts independent ETF research, comparing candidates and making informed selections without human intervention.
3. **Hybrid LLM-Heuristic Design**: A principled separation between LLM reasoning (constraint interpretation, research synthesis, explanation generation) and deterministic computation (allocation percentages, backtest metrics).
4. **Self-Validation Loop**: An LLM-as-Judge approach that evaluates recommendations against five criteria, with conditional rebalancing when concerns are identified.
5. **Three-Layer Explainability**: Console logs for developers, PDF reports for end users, and LangSmith traces for full observability.
6. **Empirical Evaluation**: Systematic comparison against three baseline LLMs across multiple investor profiles, demonstrating concrete improvements in constraint satisfaction, data grounding, and quantitative validation.

1.5 Paper Organization

Section 2 provides background on LLMs in finance, agentic AI, and portfolio theory. Section 3 reviews related work in financial agents and benchmarks. Section 4 details the system architecture and implementation. Section 5 presents empirical results comparing FinPlan-LLM against baseline approaches. Section 6 discusses broader implications and limitations. Section 7 concludes with future directions.

2 Background

2.1 Large Language Models in Finance

The application of language models to finance has evolved significantly over recent years. Early work focused on domain-specific models like FinBERT (Araci 2019), which adapted BERT for financial sentiment analysis. The emergence of GPT-class models enabled more sophisticated applications, from automated report generation to conversational financial assistants.

However, general-purpose LLMs face inherent limitations in financial applications. Their training data has a cutoff date, making them unreliable for current market conditions. They lack mechanisms to verify factual claims about specific securities. Most critically, they cannot perform real-time computations on live market data.

2.2 Agentic AI and Tool Use

The ReAct (Reasoning and Acting) paradigm (ReAct 2022) addresses some LLM limitations by interleaving reasoning traces with actions. Rather than generating complete responses in a single pass, ReAct agents iteratively think about what information they need, take actions to obtain it, observe the results, and continue reasoning.

Tool-augmented LLMs extend this capability by providing agents with external functions they can invoke. For financial applications, relevant tools might include APIs for retrieving current prices, calculating technical indicators, or accessing economic data. The LLM decides when to use each tool and how to interpret its output.

2.3 Multi-Agent Orchestration with LangGraph

LangGraph provides a framework for building complex AI workflows as directed graphs. Each node represents a processing step (potentially involving LLM calls), and edges define the flow between steps. Conditional edges enable dynamic routing based on intermediate results.

Key LangGraph concepts include:

- **State**: A shared data structure passed between nodes, accumulating information throughout the workflow
- **Nodes**: Functions that receive state, perform processing, and return updated state
- **Edges**: Connections between nodes, either unconditional or conditional based on state contents
- **Recursion Limits**: Bounds on agent iterations to prevent infinite loops

2.4 Portfolio Theory Fundamentals

Modern Portfolio Theory (MPT), introduced by Markowitz, provides the mathematical foundation for portfolio construction. Key concepts include:

Risk-Return Tradeoff: Higher expected returns generally require accepting higher risk (volatility).

Diversification: Combining assets with imperfect correlation reduces portfolio risk below the weighted average of individual asset risks.

Age-Based Heuristics: Rules of thumb like "100 minus your age in stocks" provide simple guidance for risk allocation across the lifecycle.

The system employs several quantitative metrics:

Simple Return:

$$R = \frac{P_{end} - P_{start}}{P_{start}} \times 100\% \tag{1}$$

Annualized Return:

$$R_{annual} = (1 + R_{total})^{252/n_{days}} - 1 \tag{2}$$

Volatility (Standard Deviation of Returns):

$$\sigma = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(r_i - \bar{r})^2} \tag{3}$$

Maximum Drawdown:

$$MDD = \frac{Trough - Peak}{Peak} \times 100\% \tag{4}$$

Sharpe Ratio (Risk-Adjusted Return):

$$S = \frac{R_p - R_f}{\sigma_p} \tag{5}$$

where R_p is portfolio return, R_f is risk-free rate, and σ_p is portfolio volatility. A Sharpe ratio above 1.0 indicates good risk-adjusted performance; above 2.0 is excellent.

Sortino Ratio (Downside Risk-Adjusted Return):

$$Sortino = \frac{R_p - R_f}{\sigma_d} \tag{6}$$

where σ_d considers only downside deviation.

2.5 Exchange-Traded Funds (ETFs)

ETFs serve as the primary investment vehicle in this system due to several advantages:

- **Diversification**: Single ETFs provide exposure to hundreds of underlying securities
- **Low Costs**: Expense ratios typically range from 0.03% to 0.50%
- **Transparency**: Holdings are disclosed daily
- **Liquidity**: Trade throughout the day like stocks
- **Accessibility**: No minimum investment beyond share price

3 Related Work

3.1 Financial Agent Benchmarks

Recent work has established benchmarks for evaluating financial AI agents. InvestorBench (InvestorBench 2025) provides a comprehensive framework for testing LLM-based investment agents across various tasks including portfolio allocation, risk assessment, and market timing. FinGAIA (FinGAIA 2025) focuses on agentic financial assistants, evaluating their ability to gather information, reason about financial decisions, and interact with users.

TravelPlanner (TravelPlanner 2024) from ICML 2024, while focused on travel rather than finance, established important methodological precedents for evaluating agentic systems that must satisfy multiple constraints while optimizing for user preferences.

3.2 Traditional Robo-Advisors

Commercial robo-advisors like Betterment and Wealthfront pioneered automated portfolio management. These systems typically use questionnaires to assess risk tolerance, apply mean-variance optimization to construct portfolios, and provide automated rebalancing.

However, traditional robo-advisors have limitations that motivate this work:

- **Rigid Constraints**: Limited ability to handle natural language preferences like "I want ESG but no oil companies"
- **Black Box**: Provide recommendations without detailed reasoning
- **Static Models**: Do not adapt to real-time economic regime changes
- **Limited Personalization**: Category-based rather than truly individualized

3.3 LLM-Enhanced Financial Systems

FinGPT (FinGPT 2023) demonstrated the potential of fine-tuning LLMs on financial data, achieving strong performance on tasks like sentiment analysis and financial QA. Studies of ChatGPT for portfolio construction (ChatGPT Portfolio 2023) found that while LLMs generate reasonable allocations, they lack mechanisms for verification and real-time data integration.

3.4 Agentic AI Frameworks

Several frameworks enable multi-agent AI systems. AutoGen (AutoGen 2023) provides conversational agents that can collaborate on complex tasks. CrewAI offers role-based agent orchestration. LangGraph, used in this work, provides graph-based workflow management with explicit state handling and conditional routing.

3.5 Positioning of FinPlan-LLM

Table 1 positions FinPlan-LLM relative to existing approaches:

Table 1: Comparison with existing approaches

System	NL Constraints	Real Data	Valid.	Explain.
Robo-Advisors	Limited	Yes	No	Low
Vanilla LLMs	Yes	No	No	Medium
FinGPT	Limited	No	No	Medium
FinPlan-LLM	**Yes**	**Yes**	**Yes**	**High**

4 Project Description

4.1 System Overview

FinPlan-LLM implements an 8-node LangGraph workflow organized into four phases (Figure 1):

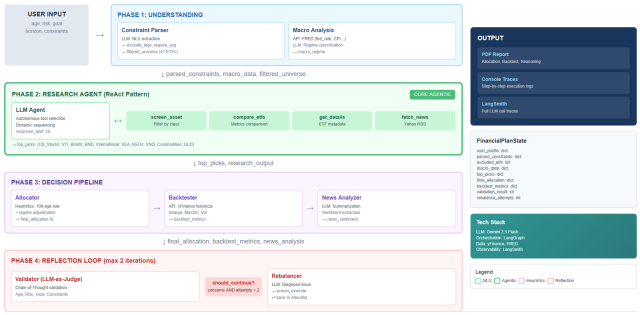


Figure 1: System Architecture: 8 nodes organized into 4 phases

Phase 1 (Understanding): Profile collection, constraint parsing, and macroeconomic analysis.

Phase 2 (Research): Autonomous ETF research via ReAct agent with tool use.

Phase 3 (Decision): Deterministic allocation, backtesting, and news analysis.

Phase 4 (Reflection): Validation with conditional rebalancing loop.

The system uses Google Gemini 2.5 Flash as its LLM backbone, selected for:

- Free API access (cost-effective for academic projects)
- Fast inference (typically under 5 seconds per call)
- Strong reasoning capabilities
- Large context window for complex prompts

4.2 Phase 1: Understanding the User

Constraint Parser The constraint parser node transforms natural language investment constraints into structured filters (Figure 2). It performs:

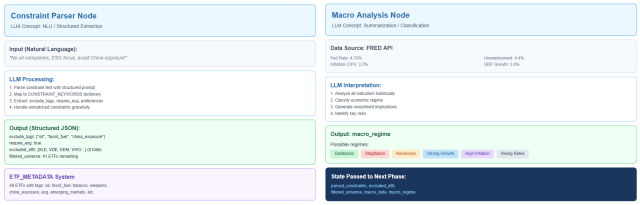


Figure 2: Phase 1: Profile collection and constraint parsing

Natural Language Understanding: The LLM interprets user constraints like "ESG focus, no oil companies" or "no tobacco, weapons, or China exposure."

Keyword Mapping: Constraints map to predefined tags in the ETF metadata (e.g., "oil" maps to tags: oil, fossil_fuel).

ETF Filtering: The system cross-references constraints against ETF metadata to identify excluded funds. For example, "no oil companies" excludes DBC (commodities), XLE (energy sector), VDE (energy), and GSG (commodities).

ESG Detection: Constraints mentioning ESG, sustainable, or responsible investing trigger ESG preference flags that boost ESG-compliant ETFs in the scoring algorithm.

Macro Analysis The macro analysis node fetches current economic indicators from the FRED API:

- **Federal Funds Rate**: The cost of borrowing that influences all interest rates. Currently used value: 3.89% (as of December 2025).
- **CPI (Inflation)**: The rate at which prices increase, eroding purchasing power. High inflation favors real assets like commodities and REITs.
- **Unemployment Rate**: A measure of labor market health. Currently: 4.4%.
- **GDP Growth**: Overall economic output measure. Currently: 3.8%.

The system classifies the economic environment into one of four regimes:

- **Goldilocks**: GDP $> 2\%$ AND Inflation $< 4\%$. Optimal conditions; favor balanced equity exposure.
- **Stagflation**: GDP $< 1\%$ AND Inflation $> 4\%$. Poor conditions; increase bonds and commodities.
- **Recession**: GDP $< 0\%$ OR Unemployment $> 7\%$. Defensive posture; maximize bonds.
- **Overheating**: GDP $> 4\%$ AND Inflation $> 5\%$. Economy running hot; reduce risk assets.

4.3 Phase 2: Research Agent

The research agent (Figure 3) implements the ReAct pattern with four specialized tools, operating with a recursion limit of 50 iterations.



Figure 3: Research Agent: ReAct pattern with 4 tools

Tool 1: screen_asset_class This tool screens ETFs within a given asset class using a quantitative scoring algorithm:

$$Score = YTD_{capped} + Sharpe_{scaled} - Expense_{penalty} + ESG_{boost} \tag{7}$$

where:

- $YTD_{capped} = \min(YTD \times 100, 30)$ (caps momentum contribution)
- $Sharpe_{scaled} = \min(Sharpe \times 20, 40)$ (emphasizes risk-adjusted returns)
- $Expense_{penalty} = ExpenseRatio \times 100$ (penalizes high fees)
- ESG_{boost}: If ESG preference enabled, multiply score by 1.2 for ESG-compliant ETFs

Worked Example: Comparing VTI vs. ESGU for US Stocks with ESG preference:

VTI: YTD 15%, Sharpe 2.3, Expense 0.03%, Not ESG-labeled

$$\begin{aligned} Score_{VTI} &= \min(15, 30) + \min(2.3 \times 20, 40) - (0.03 \times 100) \\ &= 15 + 40 - 3 = 52 \end{aligned}$$

ESGU: YTD 14%, Sharpe 2.1, Expense 0.15%, ESG-labeled

$$\begin{aligned} Score_{ESGU} &= [\min(14, 30) + \min(2.1 \times 20, 40) - (0.15 \times 100)] \times 1.2 \\ &= [14 + 40 - 15] \times 1.2 = 46.8 \end{aligned}$$

The tool returns the top candidates for further comparison.

Tool 2: compare_etfs Performs head-to-head comparison of two ETFs across 10 metrics:

1. YTD Return
2. 1-Year Return
3. 3-Year Annualized Return
4. Sharpe Ratio

5. Sortino Ratio
6. Maximum Drawdown
7. Volatility
8. Expense Ratio
9. Assets Under Management (AUM)
10. Dividend Yield

Tool 3: get_etf_details Retrieves detailed information about a specific ETF:

- Top 10 holdings with percentage weights
- Sector breakdown
- Geographic exposure
- ESG rating (if available)
- Fund characteristics (inception date, issuer, etc.)

Tool 4: fetch_market_news Fetches recent news headlines from Yahoo Finance RSS feeds. During research, this tool helps identify potential red flags (e.g., regulatory issues, management changes) that might affect ETF selection.

Important Distinction: This research tool differs from the News Analyzer node in Phase 3. The research tool provides raw headlines during ETF selection; the News Analyzer provides structured sentiment analysis for the final report.

Autonomous Behavior The agent operates autonomously without human intervention. A typical research session might proceed:

1. Agent screens US Stocks, receives top 2: VTI (score 68.2), IVV (score 59.9)
2. Agent decides to compare VTI vs IVV in detail
3. Agent reviews comparison, notes VTI has lower expense ratio
4. Agent checks VTI details to verify holdings
5. Agent selects VTI for US Stocks allocation
6. Agent repeats for remaining asset classes

4.4 Phase 3: Decision Pipeline

Portfolio Allocator The allocator node uses deterministic heuristics rather than LLM reasoning (Figure 4):

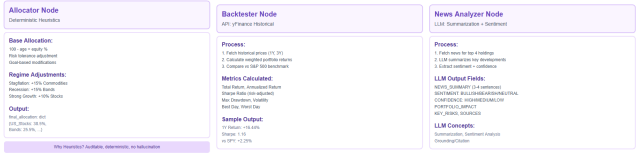


Figure 4: Allocation Heuristics: 6-step deterministic process

Step 1 - Base Allocation: Apply the "100 minus age" rule to determine baseline equity percentage. A 35-year-old starts with 65% equities.

Step 2 - Risk Adjustment: Modify based on stated risk tolerance:

- Conservative: -20% equity
- Moderate: No change
- Aggressive: $+20\%$ equity

Step 3 - Regime Detection: Classify current economic environment using macro indicators.

Step 4 - Regime Shifts: Adjust allocation based on regime:

- Goldilocks: Maintain baseline
- Stagflation: $+10\%$ commodities, $+5\%$ bonds
- Recession: $+15\%$ bonds
- Overheating: $+5\%$ commodities

Step 5 - Distribution: Allocate within asset classes (e.g., split equities between US and International).

Step 6 - Normalization: Ensure allocations sum to exactly 100%.

Backtester The backtester computes historical performance using yFinance data:

What is Backtesting? Backtesting tests an investment strategy against historical data. If you had held this portfolio over the past year or three years, how would it have performed? This is analogous to reviewing "game film" in sports.

Why Backtesting Matters:

- **Validation**: Confirms the portfolio behaves as expected
- **Risk Assessment**: Reveals actual volatility and drawdowns
- **Benchmark Comparison**: Shows performance vs. S&P 500
- **Expectation Setting**: Helps users understand realistic outcomes

Why 1-Year and 3-Year Periods? One year captures recent market conditions and momentum. Three years provides longer-term perspective while remaining relevant to current market structure.

Why Compare vs. S&P 500? The S&P 500 (SPY) serves as the standard benchmark for US equity performance. Beating SPY with similar or lower risk indicates the portfolio adds value.

Interpreting Results:

- Sharpe > 1.0: Good risk-adjusted returns
- Sharpe > 2.0: Excellent risk-adjusted returns
- Max Drawdown $< 20\%$: Acceptable for moderate risk
- Outperforming SPY: Portfolio adds value beyond passive indexing

News Analyzer The News Analyzer node runs on the top 4 portfolio holdings, providing:

- **Summary**: Concise overview of relevant news
- **Sentiment**: BULLISH, NEUTRAL, or BEARISH
- **Confidence**: LOW, MEDIUM, or HIGH
- **Portfolio Impact**: Analysis of how news affects holdings
- **Key Risks**: Identified concerns from news coverage

This differs from the research tool's news fetching: the News Analyzer always runs on final holdings, produces structured output, and appears in the user-facing PDF report.

4.5 Phase 4: Reflection Loop

Validator (LLM-as-Judge) The validator evaluates the portfolio against five criteria (Figure 5):

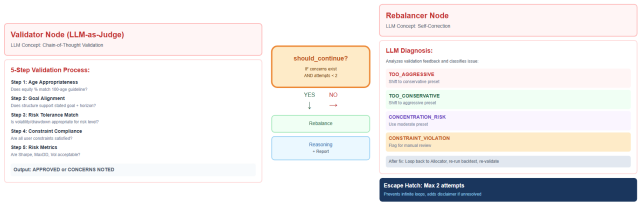


Figure 5: Reflection Loop: Validation with conditional rebalancing

1. **Age Appropriateness**: Does equity allocation match lifecycle stage?
2. **Goal Alignment**: Does strategy match stated investment goal?
3. **Risk Tolerance Match**: Is volatility consistent with risk preference?
4. **Constraint Compliance**: Are all user constraints satisfied?
5. **Backtest Reasonableness**: Are historical metrics acceptable?

The validator outputs: APPROVED (proceed to report) or CONCERNS (trigger rebalancing).

Conditional Routing The `should_continue` function implements conditional routing:

- If APPROVED: Route to report generation
- If CONCERNS and attempts $<$ max_attempts: Route to rebalancer
- If CONCERNS and attempts $\geq$ max_attempts: Route to report with disclaimer

Rebalancer When concerns are identified, the rebalancer diagnoses the issue and applies preset fixes:

- "Too much equity for age" $\rightarrow$ Shift 10% from stocks to bonds
- "Constraint violation" $\rightarrow$ Swap offending ETF for compliant alternative
- "Excessive volatility" $\rightarrow$ Increase bond allocation

Preventing Infinite Loops A critical implementation detail: the system caps rebalancing at `max_attempts = 2`. This prevents endless validation-rebalance cycles.

Why 2 attempts?

- 1 attempt: No self-correction capability
- 2 attempts: Allows one fix, preventing cascading changes
- 3+ attempts: Diminishing returns, risk of oscillation

After 2 attempts, the system exits with a disclaimer noting unresolved concerns, allowing human judgment.

4.6 Explainability Architecture

Layer 1: Console Logs (Developer) Real-time logging of each workflow step:

```
Step 1: Profile Collection
Step: Parsing Investment Constraints
  -> Parsed: {'exclude_tags': ['oil'],
              'require_esg': True}
  -> Excluded ETFs: ['DBC', 'XLE', 'VDE', '
     GSG']
Step 2: Macro Analysis
  -> LLM Interpretation: The economy...
```

Layer 2: PDF Report (End User) A professional PDF report containing (Figures 6, 7):

1. Executive Summary
2. Portfolio Allocation with pie chart
3. Constraint Analysis
4. Economic Analysis
5. Research Summary
6. Market News & Sentiment
7. Backtesting Results
8. Decision Reasoning
9. Validation Status

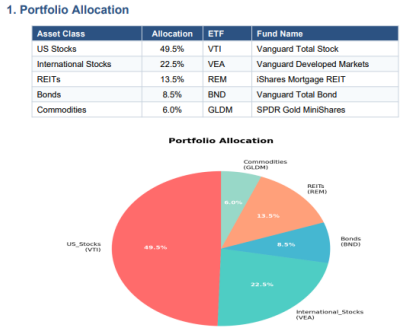
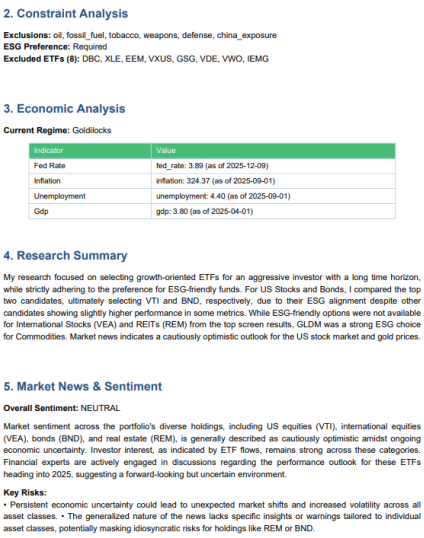


Figure 6: PDF Report: Allocation table and pie chart

Layer 3: LangSmith Traces (Full Observability) Complete workflow traces showing (Figures 8, 9):

- Every LLM call with input/output
- Tool invocations and results
- Timing for each node
- Token usage and cost



2. Constraint Analysis

Exclusions: oil, fossil_fuel, tobacco, weapons, defense, china_exposure
ESG Preference: Required
Excluded ETFs (6): DBC, XLE, EEM, VXUS, GSG, VDE, VWO, IEMG

3. Economic Analysis

Current Regime: Goldilocks

Indicator	Value
Fed Rate	fed_rate: 3.89 (as of 2025-12-09)
Inflation	inflation: 324.37 (as of 2025-09-01)
Unemployment	unemployment: 4.40 (as of 2025-09-01)
Gdp	gdp: 3.80 (as of 2025-04-01)

4. Research Summary

My research focused on selecting growth-oriented ETFs for an aggressive investor with a long time horizon, while strictly adhering to the preference for ESG-friendly funds. For US Stocks and Bonds, I compared the top two candidates, ultimately selecting VTI and BND, respectively, due to their ESG alignment despite other candidates showing slightly higher performance in some metrics. While ESG-friendly options were not available for International Stocks (VEA) and REITs (REM) from the top screen results, GLDM was a strong ESG choice for Commodities. Market news indicates a cautiously optimistic outlook for the US stock market and gold prices.

5. Market News & Sentiment

Overall Sentiment: NEUTRAL

Market sentiment across the portfolio's diverse holdings, including US equities (VTI), international equities (VEA), bonds (BND), and real estate (REM), is generally described as cautiously optimistic amidst ongoing economic uncertainty. Investor interest, as indicated by ETF flows, remains strong across these categories. Financial experts are actively engaged in discussions regarding the performance outlook for these ETFs heading into 2025, suggesting a forward-looking but uncertain environment.

Key Risks:
• Persistent economic uncertainty could lead to unexpected market shifts and increased volatility across all asset classes. • The generalized nature of the news lacks specific insights or warnings tailored to individual asset classes, potentially masking idiosyncratic risks for holdings like REM or BND.

Figure 7: PDF Report: Constraint and economic analysis

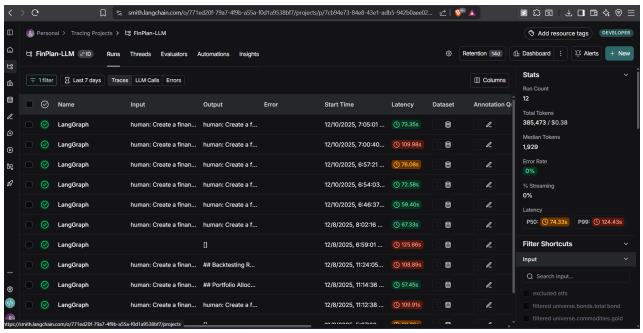


Figure 8: LangSmith: Run tracking with latency statistics

Why Explainability Matters **For Users**:

- **Trust**: Understanding reasoning builds confidence
- **Education**: Learn financial concepts through explanations
- **Accountability**: Clear attribution for decisions

For Developers:

- **Debugging**: Trace issues to specific nodes
- **Improvement**: Identify weak points for enhancement
- **Reproducibility**: Understand why outputs vary
- **Compliance**: Audit trail for regulatory requirements

4.7 User Interface

The system provides a Gradio web interface (Figure 10) with:

- Input fields for age, risk tolerance, goal, time horizon, constraints
- "Generate Plan" button to trigger workflow

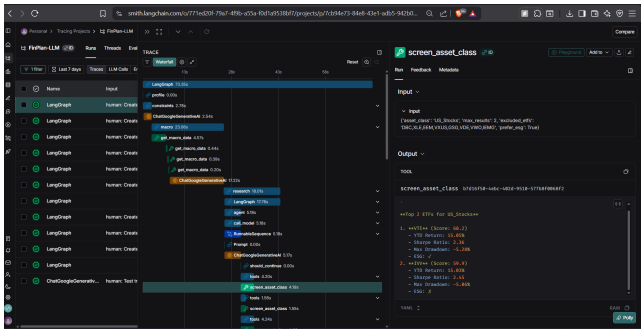


Figure 9: LangSmith: Waterfall trace showing tool outputs

- Real-time console log display
- Portfolio summary with allocation breakdown
- Downloadable PDF report
- Link to LangSmith trace

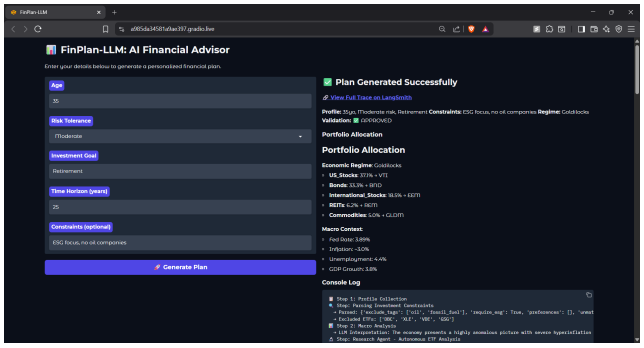


Figure 10: Gradio Interface: Input form and results display

5 Empirical Results

5.1 Experimental Setup

We evaluate FinPlan-LLM against three baseline systems:

- **Gemini 2.5 Flash**: Direct prompting (same model as our system)
- **GPT-4o**: OpenAI's flagship model
- **Claude Sonnet 4**: Anthropic's advanced model

All baselines receive identical prompts:

> "You are a financial advisor. Create a portfolio allocation for an investor with the following profile: [Age], [Risk Tolerance], [Goal], [Horizon], [Constraints]. Provide specific ETF recommendations with percentage allocations and explain your reasoning."

Test Profiles **Profile A (Moderate)**: 35 years old, Moderate risk, Retirement goal, 25-year horizon, Constraints: "ESG focus, no oil companies"

Profile B (Aggressive): 22 years old, Aggressive risk, Growth goal, 30-year horizon, Constraints: "No oil, no tobacco, no weapons, no China exposure, ESG only"

Profile C (Conservative): 62 years old, Conservative risk, Capital Preservation goal, 7-year horizon, Constraints: None

5.2 Baseline Outputs

This section presents the actual outputs from each baseline system, demonstrating what users receive from direct LLM prompting.

Profile A Baseline Outputs **Gemini 2.5 Flash:** Recommends 75% equity / 25% bonds with ESGV (45%), VSGX (30%), and EAGG (25%). Claims these ETFs "screen out fossil fuels, tobacco, weapons" and "meet ESG criteria." Suggests "Target Annual Return: 7-9%" without computation. Offers to "simulate how such a portfolio might perform" but cannot execute.

GPT-4o: Recommends ESGV (35%), ESGU (15%), EFAX (15%), ICLN (7.5%), ESGE (7.5%), EAGG (10%), plus 5% thematic. Provides detailed reasoning about ESG screening. States funds "explicitly remove fossil-fuel/reserve companies" but cannot verify current holdings.

Claude Sonnet 4: Recommends ESGV (35%), VSGX (20%), VESGX (10%), VCEB (20%), BGRN (10%), VNQ (5%). States "Target Annual Return: 7-9% long-term" and "Volatility: Moderate (12-15% standard deviation)" as specific estimates with no data source.

Profile B Baseline Outputs **Gemini 2.5 Flash:** Recommends 100% equities with ESGV (45%), DSI (15%), EFAX (20%), EMXC (10%), ICLN (10%). Notes EMXC "excludes China" to meet constraints. Suggests "Target Annual Return: 9-12%" and warns of "30-40% potential declines."

GPT-4o: Recommends ESGV (30%), DSI (15%), EFAX (20%), EMXC (10%), ICLN (10%), ESML (5%), SDG (5%), BGRN (5%). Extensive explanation of constraint compliance. Offers "comparison simulation of this ESG portfolio vs traditional S&P 500" but cannot execute.

Claude Sonnet 4: Recommends ESGU (40%), ESGD (15%), IESG (20%), IEJP (10%), ICLN (8%), VCEB (5%), VNQ (2%). States "Target Annual Return: 9-12%" and "Volatility: High (16-20% standard deviation)" without computation.

Profile C Baseline Outputs **Gemini 2.5 Flash:** Recommends 75% bonds / 20% equity / 5% cash with BND (30%), BSV (20%), VTIP (15%), VCIT (10%), SPLV (10%), VIG (10%), BIL (5%). Claims "Blended yield likely 3.5-4.5%" from training data.

GPT-4o: Recommends VGSH (20%), IGSB (20%), AGG (20%), TIP (10%), VIG (10%), USMV (10%), IDLV (5%), JPST (5%). Provides yield estimates ("∼4.7%", "∼5.0%") that may be outdated. Offers "historical backtest comparison" but cannot execute.

Claude Sonnet 4: Recommends BND (40%), IEF (20%), SCHP (10%), VIG (15%), USMV (10%), VEA (3%), VMFXX (2%). States "Target Annual Return: 4-6%" and "Volatility: Low (6-8%)" as authoritative estimates without data.

5.3 FinPlan-LLM Results

Table 2 presents FinPlan-LLM outputs with computed metrics.

Table 2: FinPlan-LLM Results Across Test Profiles

Metric	Profile A	Profile B	Profile C
Allocation	VTI 37.1% BND 33.3% EEM 18.5% REM 6.2% GLDM 5.0%	VTI 49.5% VEA 22.5% REM 13.5% BND 8.5% GLDM 6.0%	VCIT 68% GLDM 13% VOO 11% EEM 5% REM 3%
Excluded ETFs	4	8	0
Regime	Goldilocks	Goldilocks	Goldilocks
1Y Return	15.42%	17.86%	15.07%
1Y Sharpe	1.15	1.03	1.93
Max Drawdown	-10.68%	-14.28%	-4.46%
3Y Sharpe	1.20	1.17	1.46
vs SPY	+0.85%	+3.30%	+0.51%
Validation	APPROVED	APPROVED	APPROVED

Key observations from FinPlan-LLM results:

- Profile A: Systematic exclusion of 4 oil/fossil fuel ETFs (DBC, XLE, VDE, GSG) with verified ESG preference
- Profile B: Aggressive filtering of 8 ETFs including China-exposed funds (EEM, VWO, VXUS, IEMG)
- Profile C: Conservative allocation with 68% bonds achieving exceptional Sharpe ratio of 1.93 with only -4.46% max drawdown

5.4 Baseline Analysis

Common Limitations Across All Baselines All three baseline LLMs share fundamental limitations:

1. **No Real-Time Data**: Recommendations based on training data, not current market conditions
2. **Unverifiable Claims**: ETFs recommended with claims about holdings that cannot be confirmed
3. **No Validation**: No mechanism to check if recommendations meet stated constraints
4. **No Backtest**: Performance estimates are educated guesses, not computed metrics
5. **No Macro Awareness**: Generic advice regardless of current economic regime

Gemini 2.5 Flash Baseline Critique Gemini provides well-structured responses with specific ETF recommendations. However:

- Claims ESGV "screens out fossil fuels" but cannot verify current holdings
- Provides no quantitative performance metrics
- Offers to "simulate performance" but cannot execute backtests
- Does not check current Fed rate or inflation before recommending bond allocation
- ESG compliance is stated, not systematically verified

Key Gap: Gemini recommends 75% equity / 25% bonds for Profile A but provides no historical evidence this allocation achieves acceptable risk-adjusted returns.

GPT-4o Baseline Critique GPT-4o produces the most polished responses with detailed tables and extensive reasoning. However:

- Cites approximate yields ("∼4.7%", "∼5.0%") from training data that may be months out of date
- Claims EFAX has "no China exposure" but cannot verify against current fund holdings
- Offers to create "backtest comparison vs traditional 60/40 portfolio" but cannot execute
- Provides "Expected Outcomes" (4-6% return, 6-8% volatility) that are ungrounded estimates
- No awareness of current economic regime affecting allocation

Key Gap: GPT-4o's impressive formatting masks the fundamental issue that all numerical claims are unverifiable guesses.

Claude Sonnet 4 Baseline Critique Claude provides thoughtful recommendations with good explanations of reasoning. For Profile A, it suggests ESGV 35%, VSGX 20%, VESGX 10%, VCEB 20%, BGRN 10%, VNQ 5%. However:

- States "Target Annual Return: 7-9%" and "Volatility: 12-15%" without any data source
- These specific numerical estimates appear authoritative but are entirely fabricated
- Claims ESG funds "screen out oil companies" without verification capability
- No mechanism to check if current holdings match stated screening criteria
- Provides "Expected Outcomes" framing that implies computed results

Key Gap: Claude's specific numerical estimates ("7-9% return", "12-15% volatility") are particularly problematic because they appear to be computed values but are actually plausible-sounding guesses.

5.5 Side-by-Side Output Comparison

Table 3 provides a direct comparison of outputs for Profile A across all systems.

5.6 Quantitative Comparison

Table 4 compares systems across key capabilities:

Table 4: Capability Comparison: FinPlan-LLM vs. Baselines

Capability	Gem.	GPT	Claude	Ours
Real-time macro data	No	No	No	Yes
Constraint filtering	No	No	No	Yes
Computed backtests	No	No	No	Yes
Benchmark comparison	No	No	No	Yes
Self-validation	No	No	No	Yes
Explainability layers	1	1	1	3

5.7 Qualitative Comparison

Table 5 rates systems on qualitative dimensions (1-5 scale):

Table 5: Qualitative Comparison (1-5 scale)

Dimension	Gem.	GPT	Claude	Ours
Reasoning Quality	4	5	5	4
Explainability	3	4	4	5
Personalization	3	4	4	5
Data Grounding	1	1	1	5
Constraint Handling	2	2	2	5
Validation Rigor	1	1	1	5
Average	2.3	2.8	2.8	**4.8**

5.8 Ablation Analysis

Each component adds measurable value:

- **Without Constraint Parser**: System would recommend excluded ETFs (DBC, XLE) to ESG-focused users
- **Without Macro Analysis**: Allocation would ignore current economic regime, potentially over-allocating to equities during recession
- **Without Research Agent**: Would rely on static ETF list rather than dynamically evaluating current performance
- **Without Backtester**: No quantitative validation of portfolio performance
- **Without Validator**: Potentially inappropriate recommendations would reach users unchecked
- **Without News Analyzer**: Users would lack context on current market sentiment

5.9 Failure Cases and Limitations

API Dependency The system requires functioning connections to yFinance and FRED. API outages or rate limiting can cause failures.

ETF Universe Coverage The current universe of 49 ETFs may not cover all user preferences. Requests for niche exposures (e.g., "blockchain ETFs") may yield suboptimal matches.

Heuristic Simplicity The "100 minus age" rule and regime-based adjustments are simplifications of sophisticated portfolio optimization. Mean-variance optimization would provide more mathematically rigorous allocations.

Table 3: Side-by-Side Comparison: Profile A (35yo, Moderate, ESG, No Oil)

Aspect	Gemini 2.5	GPT-4o	Claude Sonnet 4	FinPlan-LLM
Allocation	ESGV 45%, VSGX 30%, EAGG 25%	ESGV 35%, ESGU 15%, EFAX 15%, ICLN 7.5%, ESGE 7.5%, EAGG 10%	ESGV 35%, VSGX 20%, VESGX 10%, VCEB 20%, BGRN 10%, VNQ 5%	VTI 37.1%, BND 33.3%, EEM 18.5%, REM 6.2%, GLDM 5.0%
Constraint Handling	Verbal claim: "screens out fossil fuels"	Verbal claim: "explicitly remove fossil-fuel companies"	Verbal claim: "screen out oil companies"	Systematic: 4 ETFs filtered (DBC, XLE, VDE, GSG)
Performance Estimate	"Target: 7-9%" (no source)	"Expected: similar to market" (vague)	"Target: 7-9%, Vol: 12-15%" (no source)	1Y Return: 15.42%, Sharpe: 1.15 (computed)
Macro Awareness	None	None	None	Goldilocks regime detected via FRED
Validation	None	None	None	LLM-as-Judge: APPROVED

Agent Variability The ReAct agent's autonomous behavior can lead to different research paths on identical inputs. While generally producing similar final selections, the specific comparison sequence may vary.

News Analysis Limitations Yahoo Finance RSS provides limited, general news. Specialized financial news APIs would enable more targeted analysis.

5.10 Performance Characteristics

- **End-to-End Latency**: 60-110 seconds
- **Token Usage**: ∼385,000 tokens per run
- **Cost**: ∼$0.38 per query (Gemini pricing)
- **LLM Calls**: 8-15 depending on research depth

6 Broader Implications

6.1 Democratization of Financial Planning

FinPlan-LLM demonstrates that sophisticated financial planning can be made accessible:

- **Cost**: $0.38 per comprehensive plan vs. hundreds of dollars for human advisors
- **Accessibility**: Available 24/7, no minimum account size
- **Personalization**: Natural language constraints rather than rigid questionnaires
- **Education**: Detailed explanations teach users about their investments

6.2 Risks and Limitations

- **Not Professional Advice**: Cannot replace licensed financial advisors for complex situations
- **Market Risk**: Historical backtests do not guarantee future performance
- **Algorithmic Bias**: Heuristics embed assumptions that may not suit all users
- **Technical Failures**: API dependencies create potential points of failure

6.3 Societal Considerations

- **Financial Literacy**: May improve understanding of investment concepts
- **Market Effects**: Wide adoption could influence ETF flows
- **Regulatory Implications**: Automated advice systems face evolving regulatory frameworks
- **Employment**: May affect demand for entry-level financial advisors

7 Conclusions

7.1 Summary of Contributions

This paper presented FinPlan-LLM, a multi-agent financial advisory system that addresses fundamental limitations of vanilla LLM-based financial advice. The system demonstrates that combining LLM reasoning with deterministic heuristics produces recommendations that are both personalized and verifiable.

7.2 Key Takeaways

1. **Hybrid Architecture**: The principle of "LLMs for reasoning, heuristics for hard decisions" provides a template for domains where accuracy matters
2. **Explainability is Essential**: Three-layer transparency (console, PDF, traces) serves different stakeholders
3. **Agentic Patterns Enable Autonomy**: ReAct with tools allows sophisticated research without human intervention
4. **Escape Hatches Required**: Bounded iteration (max 2 rebalancing attempts) prevents infinite loops while allowing self-correction

7.3 Future Work

- **Additional Data Sources**: Integrate Bloomberg, Refinitiv, or alternative data

- **Mean-Variance Optimization**: Replace heuristics with mathematical optimization
- **Multi-Turn Conversation**: Enable iterative refinement of recommendations
- **User Feedback Loop**: Learn from user acceptance/rejection of recommendations
- **Broader Asset Classes**: Support individual stocks, options, and alternative investments
- **Real-Time Monitoring**: Track portfolio performance and alert on rebalancing needs

7.4 Advice for Future Students

For students undertaking similar projects:

1. **Start End-to-End**: Build a minimal working pipeline before optimizing components
2. **Invest in Observability**: LangSmith traces saved hours of debugging
3. **Respect the Domain**: Financial decisions require deterministic reliability, not just plausible outputs
4. **Embrace Hybrid Design**: LLMs and algorithms each have strengths; combine them thoughtfully
5. **Plan for Failures**: Escape hatches and graceful degradation matter in production systems

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Code Availability

The complete implementation along with the output reports for the 3 forementioned cases are available at: https://github.com/mamaniaharsh7/FinPlan-LLM

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A LLM Concepts Demonstrated

Table 6 maps LLM concepts from the course to system components:

Table 6: LLM Concepts in FinPlan-LLM

Concept	Component
Prompt Engineering	All LLM nodes
Tool Use	Research Agent (4 tools)
ReAct Pattern	Research Agent
Multi-Agent Systems	LangGraph workflow
LLM-as-Judge	Validator node
Structured Output	Constraint Parser
Chain-of-Thought	Reasoning traces
Retrieval (RAG-like)	ETF metadata lookup
State Management	LangGraph state
Conditional Routing	Validation loop
Observability	LangSmith integration

B ETF Universe

The system includes 49 ETFs across 5 asset classes:

US Stocks (12): VTI, VOO, IVV, SPY, QQQ, VUG, VTV, SCHD, VIG, ESGU, ESGV, SUSA

International Stocks (10): VEA, IEFA, VWO, EEM, VXUS, IXUS, IEMG, ACWI, VSGX, ESGD

Bonds (12): BND, AGG, VCIT, VCSH, VGSH, TLT, IEF, LQD, HYG, BNDX, USIG, EAGG

REITs (7): VNQ, SCHH, IYR, REM, REET, VNQI, SRET

Commodities (8): GLD, GLDM, IAU, SLV, DBC, GSG, PDBC, XLE

C Sample Console Output

Listing 1: Console output for Profile A

```
Step 1: Profile Collection
Step: Parsing Investment Constraints
  -> Parsed: {'exclude_tags': ['oil',
      'fossil_fuel'], 'require_esg': True}
  -> Excluded ETFs: ['DBC','XLE','VDE','GSG
     ']

Step 2: Macro Analysis
  -> Fed=3.89%, Inflation=3.0%,
     Unemployment=4.4%, GDP=3.8%
  -> Regime: Goldilocks

```

```
Step: Research Agent - Autonomous ETF
    Analysis
  -> Researching 5 asset classes...
  -> Final picks: {'US_Stocks': 'VTI',
      'International_Stocks': 'EEM',
      'Bonds': 'BND', 'REITs': 'REM',
      'Commodities': 'GLDM'}

Step: Portfolio Allocation
  -> US_Stocks: 37.1% -> VTI
  -> Bonds: 33.3% -> BND
  -> International_Stocks: 18.5% -> EEM
  -> REITs: 6.2% -> REM
  -> Commodities: 5.0% -> GLDM

Step: Backtesting Portfolio
  -> 1Y Return: 15.42%, Sharpe: 1.15
  -> vs SPY: +0.85% outperformance

Step: Portfolio Validation
  -> APPROVED

PDF Report saved: financial_plan_report.pdf
```